

FFAR Final Progress Report

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	552176
Award Program	
Project Title	Stolen Kiss – an advanced breeding solution for the animal welfare trait of genetic castration.
Grant Start Date	01/02/2018
Grant End Date	08/15/2021
Total Grant Budget	\$514,643.00
Total FFAR Budget	\$499,443.00
Matching Funder(s)	This proposal type did not require matching funds. However, DNA Genetics did provide in-kind support for this project in 2018 by providing cull sows as a female germplasm source to make edited embryos to transfer to recipient gilts.
Final Report Date	01/28/2022
Project Director/Principal Investigator Information	
Full Name	Tad Sonstegard
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Grantee Organization Information	
Organization Name	Recombinetics
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Authorized Signing Official (ASO) Information	
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ASO Title	CEO
Email	tad@acceligen.com
Phone	612 979-8367



General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

St. Paul, MN	CD-4 th
Glencoe, MN	CD-7 th
Royalton, MN	CD-6 th
Eagan, MN	CD-2 nd
Clay Center, NE	CD-3 rd
Lincoln, NE	CD-1 st
Oneida, SD	CD-at large

- 1.2. How many new jobs were created because of this grant funding?

One new position was created. This position serially funded a Master's student and then was switched to a Senior Research Associate at Recombinetics. Additionally, through this project and internal funding by Acceligen, a graduate student at the University of Lincoln, NE was financially supported to execute the work needed to support fertility rescue trials completed in 2021.

- 1.3. How many jobs were maintained because of this grant funding?

At least one full FTE was maintained throughout most of the grant period.

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No change – Acceligen is still a wholly owned subsidiary of Recombinetics.

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.



Male piglets used for pork production are routinely castrated to prevent “boar taint” and pig to pig and pig to human aggression. Boar taint affects the marketability of meat derived from uncastrated male carcasses. Currently, castration is done surgically and is a concern due to negative impact on animal well-being. To investigate alternatives to surgical castration, we bred experimental pigs were born with a gene-edited alteration to stop expression of a protein that triggers the release of hormones necessary for sexual maturation. The protein is expressed from a gene called *KISS1*, which is normally expressed in the pig’s pituitary gland during puberty, triggering the production of hormones which propel puberty forward. A total of 684 edited embryos produced 69 live piglets, of which 26 animals had transmissible genomic alterations in *KISS1* to eliminate its function. Some of these founder animals displayed lack of gonadal development, regardless of sex. Animals carrying both edited and unedited versions of *KISS1* were mated. Two rounds of breeding produced animals for testing boar taint and fertility rescue. Overall, the progeny of the founder animals went through a complete regiment of physical assessment that included quantification of pituitary and gonadal hormones, boar taint compounds, testicular volume, and daily gain of weight to show the effects of the gene edited *KISS1*. For fertility rescue, our results showed that the treatment of gilts with common commercial products or a *KISS1* analog did not induce puberty. Fat samples tested for skatole and androsterone showed edited animals had statistically significant different levels of androsterone, which were below the threshold necessary for production of boar taint compounds. A second round of female fertility rescue is still in progress. In conclusion, gene editing to disrupt *KISS1* expression can stop boar taint production and cause sterility in both males and females by inhibiting puberty. Additional studies are needed to properly induce fertility from these *KISS1* KO genotypes. Once these can be achieved in a commercially acceptable way, then further studies will be need to examine the change in growth rate and meat quality.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

First, we demonstrated the feasibility of using commercial lines of pigs to make gene-edited offspring. This practical outcome was considered a major achievement for our two commercial partners in this project.

Next, the reproductive biology of female animals (gilts) is not well

understood relative to herd longevity and lifetime productive fecundity. Future lines of KISS-/- gilts, which do not express functional KISSPEPTIN (protein), provide an excellent experimental model to better understand the endocrine communication between the brain and the ovary before and during puberty. As such, future reproductive studies on KISS-/- animals could lead to better management practices and animal well-being for sows and gilts.

Our other major finding was the validated inhibition of boar taint production in KISS-/- market pigs. This finding supports potential use of gene editing to make lines of pigs that produce piglets with improved animal well-being. However, additional studies are needed to further elucidate commercially viable methods to rescue sterile males and females for nucleus herd breeding.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;

The most impactful finding was to demonstrate that progeny of gene edited parents, which did not express functional KISSPEPTIN (protein), had statistically significant lower levels of boar taint components than their contemporary littermates that had one or two functional copies of *KISS1* (gene). This result shows that the main goal of genetic castration can be achieved through application of gene editing technology. The issues around making the trait commercially feasible through results from fertility rescue and feed efficiency are still unresolved.

- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and

The reproductive biology of female animals (gilts) is not well understood relative to herd longevity and lifetime productive fecundity. The female progeny, which did not express functional KISSPEPTIN (protein), provide an excellent experimental model to better understand the endocrine communication the brain and the ovary before and during puberty. As such, the animals produced in this project will become part of ongoing studies to unravel these physiological pathways, which could lead to better management practices and animal well-being for sows and gilts. This was an unexpected outcome from this project.

- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.
 - Nothing to report

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Specific Goal 1. Implement a method for TALEN-mediated, reversible genetic castration based on targeted inactivation of porcine KISS1 (kisspeptin).

This goal was completed 100% in making 26 founder animals with genetic edits in KISS1. The milestone completion slated for completion in March 2019 was actually completed in November 2019.

Specific Goal 2. Through animals produced by founder matings (KISS1+/-), characterize growth to market weight across the various progeny genotypes (i.e. KISS1+/+ vs. KISS1+/- and KISS1-/-).

This goal was completed but not to support the rigor of statistical significance necessary to support peer-review publication. The objective was completed in August 2021.

Specific Goal 3. Through animals produced by F1 matings (KISS1+/-), test infertility reversal by characterizing sexual development and reproductive parameters of KISS1-/- genotypes undergoing ectopic, wave and pulsatile ligand treatments that target GPR54 activation or similar regimens of follicle-stimulating hormone (FSH) and luteinizing hormone (LH) therapies.

Two fertility trials from founder progeny were completed, one in 2020 (female) and one in 2021 (male). A second round of breeding was done to produce a second cohort of females for

fertility rescue. This study is ongoing until May 2022.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

The main objectives 2 and 3 were switched in order through consultation with our industry partners and approval by our FFAR program officer. Fertility rescue was essential to prove before industry funding for a relatively large feed study would be provided (900 animals to reach statistical significance). There were no obvious differences in carcass weight for gene edited animals, but because fertility rescue was not achieved there is no statistical support for this observation.

CRISPR-Cas9 was used instead of TALENs to produce the gene edits.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Project summary. Most male pigs are surgically castrated to avoid puberty-derived boar taint and aggressiveness. This surgical intervention is a main welfare issue in the swine industry. To generate castration-free (CF) pigs, we have used a genome editing methodology for disrupting swine *KISS1* and thus keep pigs in a prepubertal state by inducing hypogonadotropic hypogonadism (HH). Herein, the first *KISS1*-edited large animals were generated via microinjection of CRISPR/Cas9-ribonucleoproteins (RNPs) and single-stranded donor oligonucleotides (ssODNs) into porcine zygotes. Four highly-edited pigs for *KISS1* were KO as they resembled the HH phenotype and failed to reach sexual maturity. In these animals, phenotypical abnormalities unrelated to the reproductive system were not observed. Forty liveborn pigs bore *KISS1*-disrupted alleles at different frequencies and did not phenotypically differ from their WT counterparts. From this founder generation (F0), 51, 60, and 36 non-mosaic F1 pigs of genotypes *KISS1* +/+ (WT), *KISS1* +/- (heterozygous), and *KISS1* -/- (KO) were produced, respectively. Avoiding mosaicism was necessary to obtain clean phenotypes. It allowed observing that CF pigs (*KISS1* KO) have considerably reduced compounds related to boar taint. Also, this was of relevance to initiate the trials to restore fertility in *KISS1* -/- individuals, which remains elusive. If the fertility of kisspeptin deficient boars and gilts can be restored, preventing pigs from advancing through puberty by disrupting *KISS1* holds the potential to replace surgical castration in male piglets.

Generation of *KISS1*-edited pigs using single guide RNAs (sgRNAs)/Cas9 RNPs and



ssODNs. To introduce loss-of-function mutations in swine KISS1, nine sgRNAs were designed to target the second exon of this gene, near or preceding the C6 core functional peptide. After transfecting these sgRNAs in porcine fibroblasts, the two gRNAs with the highest mutation activity were tested in pig parthenotes via microinjection. Alongside the sgRNA, homology-directed repair (HDR) templates HDRstop codon (SC) and HDRblocking mutations (BM) were also microinjected. This was to increase the odds of inducing specific edits capable of disrupting KISS1 (HDRSC) and/or maintaining KISS1 function (HDRBM). Further optimizations allowed selecting one sgRNA and determining the adequate concentration of CRISPR/Cas9 reagents for high heterozygosity rates and low rates of bi-allelic mutation. Then, these editing tools were deployed to Duroc and Yorkshire pigs from commercial genetics lines, via microinjection of in vivo fertilized single-cell embryos. A total of 684 intracytoplasmically injected zygotes were transferred into 19 recipient females. After full-term pregnancies, 69 (35 male - 34 female) liveborn piglets were farrowed in nine litters. For decoding the genotypes, a region surrounding the predicted CRISPR cut site was amplified by polymerase chain reaction and submitted for Sanger and next-generation sequencing. Overall, 26 animals were generated with KISS1-disruptive editing efficiencies between 30% and 90%, and in eight pigs this rate was greater than 90%.

Phenotypic analysis of mosaic KISS1-edited pigs. From the F0 population, four pigs displayed hypogonadism and were classified as KISS1 KO. Remarkable reduced testicular development was observed in the only two alive male pigs whose KISS1-disruptive efficiencies were 100%. Two highly-edited gilts presented considerably smaller ovaries and uteri, as well as the absence of follicles, when compared to other necropsied females. Pigs of both sexes that displayed hypogonadism failed to go through puberty, and no phenotype abnormalities other than underdevelopment of the genitalia were observed. The other generated KISS1-mosaic edited and WT pigs reached sexual maturity and were phenotypically normal based on macroscopic examinations. Regarding the testicular histology of F0 pigs, cell markers in KISS1 KO boars showed that germ and Sertoli cells are reduced in number while the latter and tubules are small. However, the observed histological structures suggest that it may be possible to induce spermatogenesis.

Hormonal profiling of mosaic F0 pigs. In boars, the KISS1-disruptive efficiency >90% group had substantially less luteinizing hormone (LH) secretion throughout development. There were separate effects of age ($P < 0.0001$) and KISS1-disruptive efficiency group ($P < 0.01$). A clear effect of age on serum concentrations of LH was observed in female pigs ($P < 0.001$). Although average serum concentrations of testosterone in development were numerically less in the group containing KISS1 KO animals, this difference did not reach significance due to high variance among the groups; the only effect detected was age ($P < 0.0001$). These findings agree with the predicted hypogonadotropism in animals that had already shown hypogonadism, both typical of KISS1-deficient individuals. Currently, estradiol, progesterone, and follicle stimulating hormone levels are being evaluated.

Production of the F1 generation of pigs. Previous evaluations enabled the selection of optimal F0 pigs to be used for breeding stock. By implementing artificial insemination with F0 boars and gilts, two rounds of mating were done and F1 liveborn pigs were obtained in the following proportions: 51 KISS1 +/+, 60 KISS1 +/-, and 36 KISS1 -/-. Phenotypic data of F1 animals were consistent with our observations in F0 mosaic KISS1-edited pigs. KISS1 KO pigs developed significantly smaller gonads than their heterozygous or WT counterparts. F1 KISS1-disrupted pigs were also phenotypically normal except for hypogonadotropic hypogonadism and subsequent infertility. Nevertheless, assessment of pig growth performance with a

statistically significant population is required for being conclusive, which is beyond the scope of this phase of the project.

Trials to restore fertility in KISS1 KO F1 pigs. Two attempts to reestablish fertility have been carried out, and there is a third one in progress. In the 1st trial with females, 5 KISS1 +/+, 4 KISS1 +/-, and 4 KISS1 -/- gilts were treated with commercially available products, including superovulation protocols and ovulation inducers. WT and heterozygous gilts presented estrous and ovulated, but KISS1 KO females did not respond to this approach. Ovarian histological sections suggested that there might be some differences in the ovaries of KISS1 +/- and KISS1 +/+ gilts. There were fewer observable follicles in heterozygous than in WT females, but it did not prevent KISS1 +/- gilts from being fertile. The ovaries of the KISS1 KO gilts under the microscope showed gonadotropin response in the follicles. They were able to stimulate follicular growth in those animals because they had antral follicles. It suggested that follicular growth may be stimulated with other treatments besides the standard approaches. To further explore this, in the 2nd fertility rescue attempt with gilts, C6 is being used besides other complementary hormone compounds. For this 2nd experiment of females, 5 WT, 6 heterozygous and 4 KO animals are being used. Regarding the boars, 11 males (3 KISS1 +/+, 4 KISS1 +/-, and 5 KISS1 -/-) were treated with C6, but there were no behavioral changes driven by testosterone, and semen collection was not achieved. Considering that in early life male pig Sertoli cells undergo differentiation and proliferation, at different moments, adjustments about the timing of C6 treatments may increase the likelihood of rescuing fertility in boars.

Androstenone and skatole concentrations in KISS1 KO boars. Before and after the C6 treatment of F1 boars, fat samples were collected to assess the levels of the main compounds associated with boar taint. Results suggest that differences in the androstenone and skatole levels were statistically significant between KISS1 -/- and KISS1 +/+ or KISS1 +/- animals before treatment (Sig. <0.05). Androstenone levels after treatment were statistically significantly different from their initial levels in KISS1 KO pigs (Sig. 0.013). In all other cases, there were no significant differences before and after treatment. Although a larger sample size would be ideal to have enough statistical precision and power to draw conclusions, the numeric reduction of androstenone and skatole levels was clear in KISS1 KO boars. Therefore, it is not expected to detect boar taint in KISS1-disrupted male pigs.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

Statistical analyses were used to examine differences between all groups of genotypes with a cohort of animals using approaches that can meet peer-review standards. All animal experiments were approved by IACUC committee and the scope of animal numbers was only limited by numbers of animals available for each genotype.

- 2.8. List key stakeholders that could be served by results of this project.

Swine Researchers (including those academic and company members of the National Swine Improvement Federation and EFFAB in the EU), Swine Genetic Companies, and National Pork Board

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

Acceligen and Hendrix provided significant additional funding (>\$100,000) to support animal housing, nutrition, semen collection, and phenotyping.

Covid 19 closures of USG facilities severely delayed completion of phenotyping of hormone levels in experimental pigs (still ongoing). It also delayed synthesis of C6 in France necessary for testing fertility rescue.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Nothing to report



- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

This project supported at various times – one Animal Science Master’s student, and two different Ph.D. students in animal reproduction.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Conference Abstracts, Popular press articles, scholarly publications (in prep.), images of gene edited pigs (used for phenotyping)

- 3.2. Please provide a list of citations for the information products produced during the project.

Flórez J, Martins K, Solin S, Bostrom J, Ganbaatar U, Kern D, Coutts A, Ongaratto F, Rodríguez P, Larson S, Lents C, Huisman A, Rathje T, Carlson D, and ad Sonstegard T. Generation of castration-free swine by CRISPR/Cas9-mediated knockout of the gene KISS1. Poster at: Genome Writers Guild Conference; 2019; Ames, IA.

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Flórez JM, Martins K, Solin S, Bostrom JR, Rodríguez P, Ongaratto F, Larson SA, Ganbaatar U, Coutts AW, Kern D, Kim ES, Huisman A, Carlson DF, Murphy TM, Lents CA, and Sonstegard TS. CRISPR/Cas9-editing of KISS1 to generate pigs with hypogonadotropic hypogonadism as a castration free trait and a mosaic parental line. 2022. Frontiers Livestock Genetics (in review). Special Issue on Genetic Engineering in Animals.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not

necessary to include the publications already specified above.

<http://www.acceligen.com/newsroom/>

<https://www.nationalhogfarmer.com/livestock/castration-solution>

<https://www.pigprogress.net/Health/Articles/2018/3/Can-gene-editing-solve-the-castration-issue-252387E/>

<https://www.nationalhogfarmer.com/livestock/update-gene-editing-instead-physical-castration>

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

Nothing to report.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Nothing confidential to report related to gene editing and pig breeding.

The C6 compound used for fertility rescue trials was obtained through a Material Transfer Agreement between USDA and INRA (France), so terms of that agreement must be followed regarding use of the C6 compound.

We expect all results generated to date, even those with C6, will be reported in two different peer-reviewed journal articles. The first two papers (paper 1 in review) are: 1) CRISPR/Cas9-mediated generation of pigs harboring KISS1-disrupted alleles and recapitulating hypogonadotropic hypogonadism and 2) Circulating concentrations of gonadotropins and gonadal steroids during



development in pigs depends on the number of functional KISS1 alleles. Other titles (delayed by Covid rules for USG workers) will be: 1) Interrelationship of KISS1 alleles, gonadotropin-releasing hormone stimulation and circulating concentrations of gonadotropins and testosterone in boars, 2) Pituitary and gonadal function of KISS1 knockout gilts; effects of KISS1 copy number and stimulation with gonadotropin-releasing hormone and serum gonadotropins, 3) Pituitary and gonadal response of pigs to the kisspeptin receptor agonist C6 is related to the number of functional KISS1 alleles, and 4) A functional KISS1 allele is required for neurokinin B-stimulated secretion of gonadotropin hormones in pigs.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

Executing a cooperative research agreement with USDA, ARS and University Nebraska to maintain lines of pigs for future study. All other data and meta-data will be available through published peer-reviewed manuscripts (yet to be submitted).

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

The data generated from this project will be available in peer-reviewed journal articles. They also are available by request to Acceligen by contacting the project PI, Tad Sonstegard. All the unpublished data (series of photographic images of phenotypes) will be made available from the Acceligen database after review of the data request.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

The data sets are very similar due to the repeated phenotypic measures between the 3 different cohorts of pigs bred in this project see figure below in 4.3.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if



necessary, in a compressed folder.

F0	F1 Gen 1st round	F1 Gen 2nd round
Genotype WT 18 <30% 9 80-10% 16 70-10% 7 >90% 5	Genotype WT 39 HET 18 KO 16	Genotype WT 11 HET 12 KO 18
Phenotype Live Animals - 40-220d * Photo: females & males * Blood: females & males * Weight: females & males * Testicular volume Dead Animals * Photo of reproductive tract * Weight ovaries & testicles * Ovaries follicles * Testicles measurements * Ovaries / testes snap frozen and histology	Phenotype Live Animals - 40-220d * Photo: females & males * Blood: females & males * Weight: females & males * Testicular volume * Libido Assessment - FR group Dead Animals - 50d * Back fat / photo - boars * Reproductive tract - gilts N=2	Phenotype Live Animals * None Dead Animals - 50d * Females none * Males - Back fat - Presence of testicles (some)
Breeding Screen Collection from HET F1 - 1st round F1 - 2nd round	Fertility Rescue Live Animals - F & M * Blood for Hormonal assay Dead Animals * Reproductive tract - F * Ovaries for histology * Follicular Count * Last Rib Backfat - M * Body Weight - M * Testicular / Epididymis measurement * Backfat biopsies for boar carc - M	Fertility Rescue Live Animals - F * Blood for Hormonal assay Dead Animals * Reproductive tract - F * Ovaries for histology * Follicular Count